IMT Atlantique
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OLA 2021
International Conference
on
Optimization and Learning

A Learning-based Iterated Local Search Algorithm for solving the Traveling Salesman Problem

Maryam KARIMI-MAMAGHAN
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Mehrdad MOHAMMADI
Bastien PASDELOUP

OUTLINE

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BACKGROUND

PROPOSED ALGORITHM

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RESULTS

CONCLUSION

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Introduction

Combinatorial Optimization Problems (COPs)

Meta-heuristics (MHs)

Machine Learning (ML)

Introduction

Combinatorial Optimization Problems (COPs)

- Discrete decision variables and finite search space

Meta-heuristics (MHs)

Machine Learning (ML)

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Combinatorial Optimization Problems (COPs)

- Discrete decision variables and finite search space

Meta-heuristics (MHs)

- Good substitutes for exact algorithms

Machine Learning (ML)

Introduction

Combinatorial Optimization Problems (COPs)

- Discrete decision variables and finite search space

Meta-heuristics (MHs)

- Good substitutes for exact algorithms

Machine Learning (ML)

- Used **solely**/ **integrated** into MHs for solving optimization problems
- ML techniques can be integrated into MHs for different purposes

Introduction

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journal homepage: www.elsevier.com/locate/ejor

Invited Review

Machine learning at the service of meta-heuristics for solving combinatorial optimization problems: A state-of-the-art

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Operator Selection

- MHs use Local search and perturbation operators
- Non-stationary solution space
- Availability of many general and problem-specific operators
- Operators with different characteristics

Introduction

Operator Selection

- MHs use Local search and perturbation operators
- Non-stationary solution space
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Operator selection strategies

  - Pre-determined order
  - Random selection
  - Adaptive operator selection

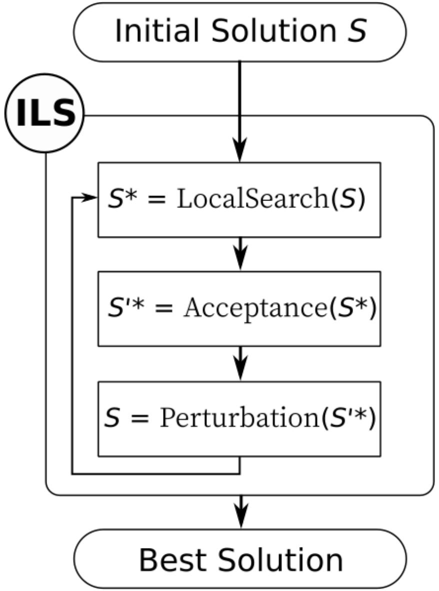
Introduction

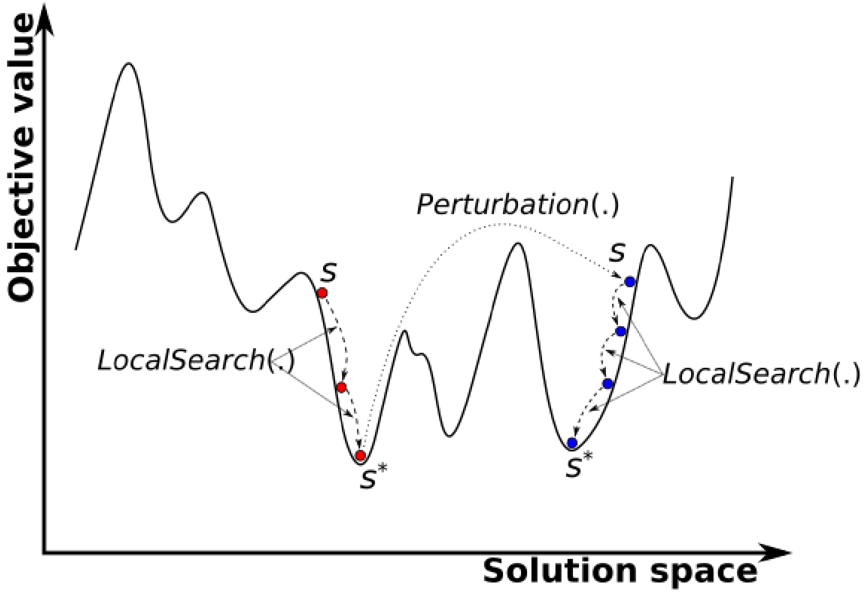
This work

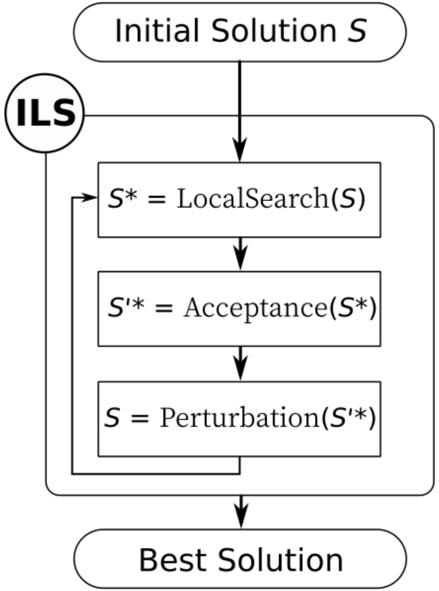
- Integrate Q-learning into a the **iterated local search** meta-heuristic for solving **combinatorial optimization problems**
- Purpose of integration is **adaptive operator selection**
- Present an application to the **traveling salesman problem**

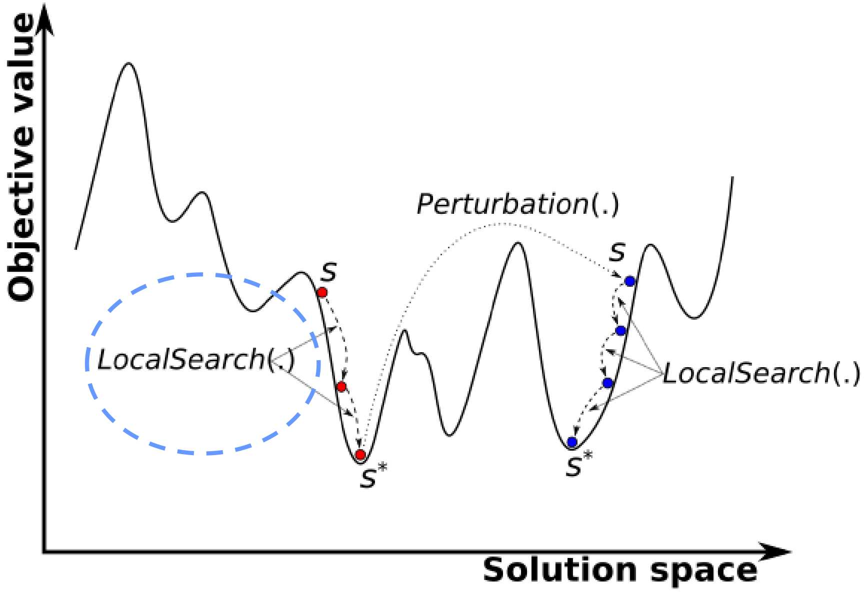
Background

Iterated Local Search (ILS)

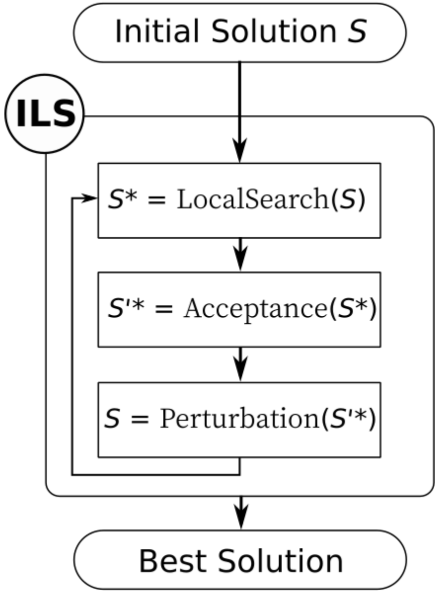


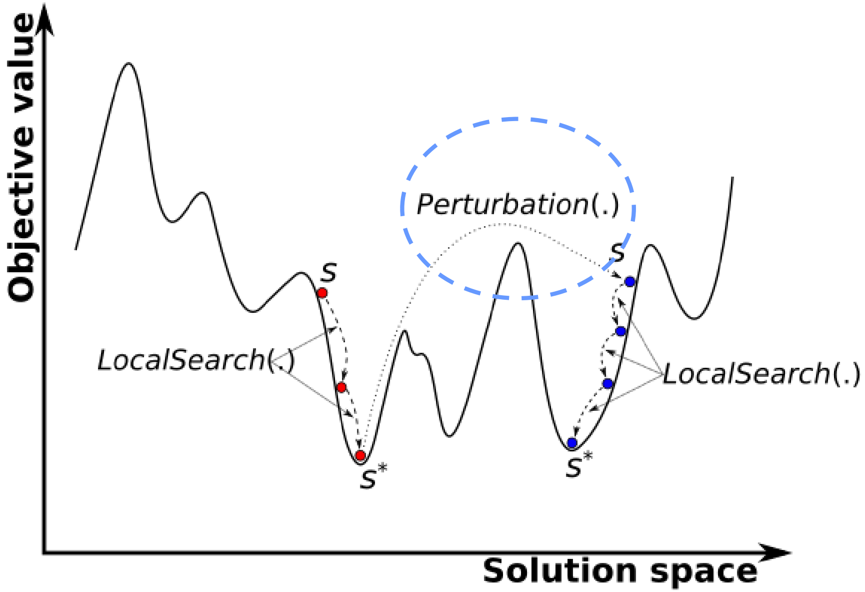



Initial Solution S
ILS
$S^* = \mathrm{LocalSearch}(S)$
$S'^* = \mathrm{Acceptance}(S^*)$
$S = \mathrm{Perturbation}(S'^*)$
Best Solution


Objective value
Solution space
Perturbation(.)
LocalSearch(.)
LocalSearch(.)
S
S^*
S
S^*

Iterated Local Search (ILS)

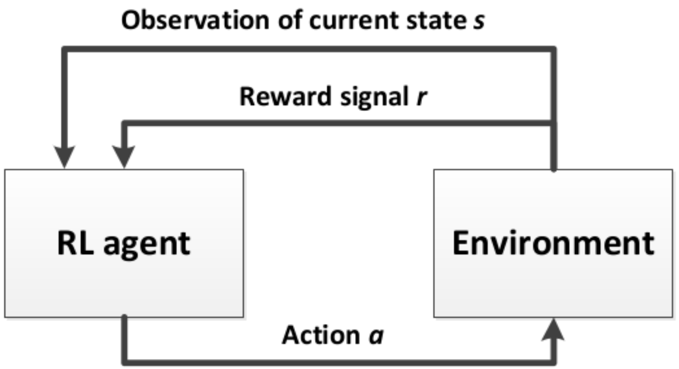




The Q-Learning algorithm

- A reinforcement learning algorithm
- Model-free algorithm
- Define states, actions, and Q-value for each (state, action)
- Bellman equation

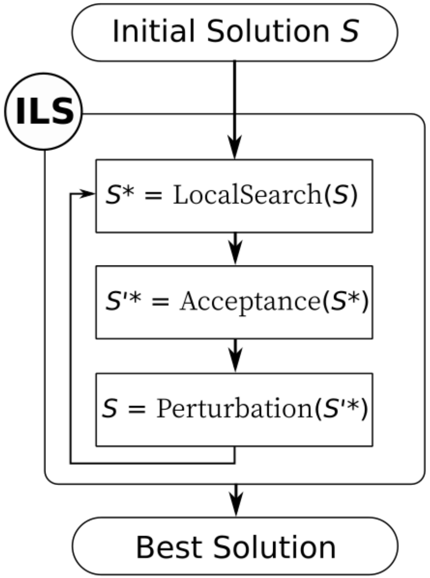
$$Q(s,a) = Q(s,a) + \alpha[r + \gamma . max_{a'} Q(s', a') - Q(s,a)]$$



Wauters, et al., 2013. Boosting metaheuristic search using reinforcement learning. In Hybrid Metaheuristics (pp. 433-452). Springer, Berlin, Heidelberg.

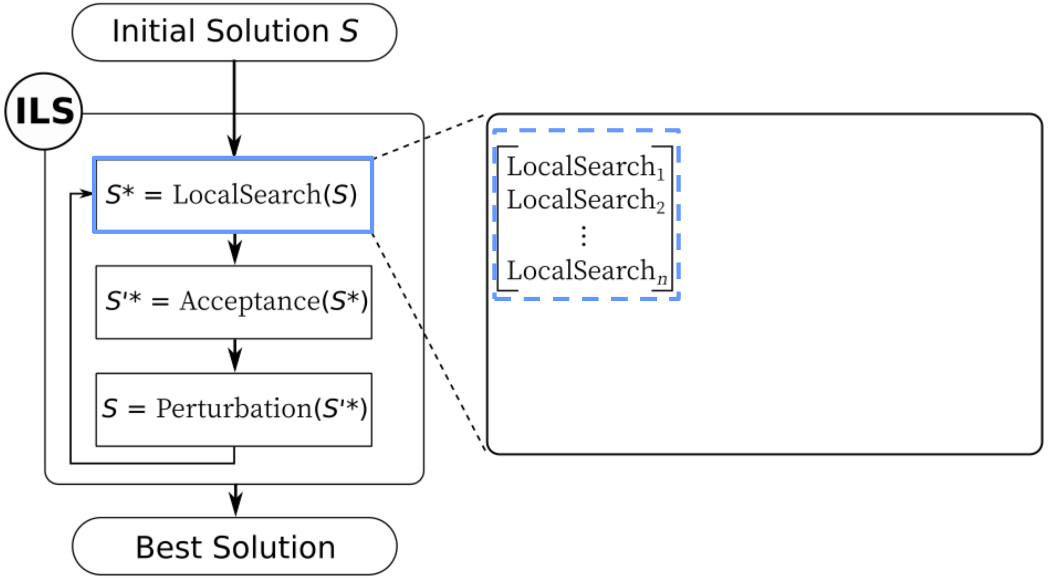
Proposed algorithm

Iterated local search

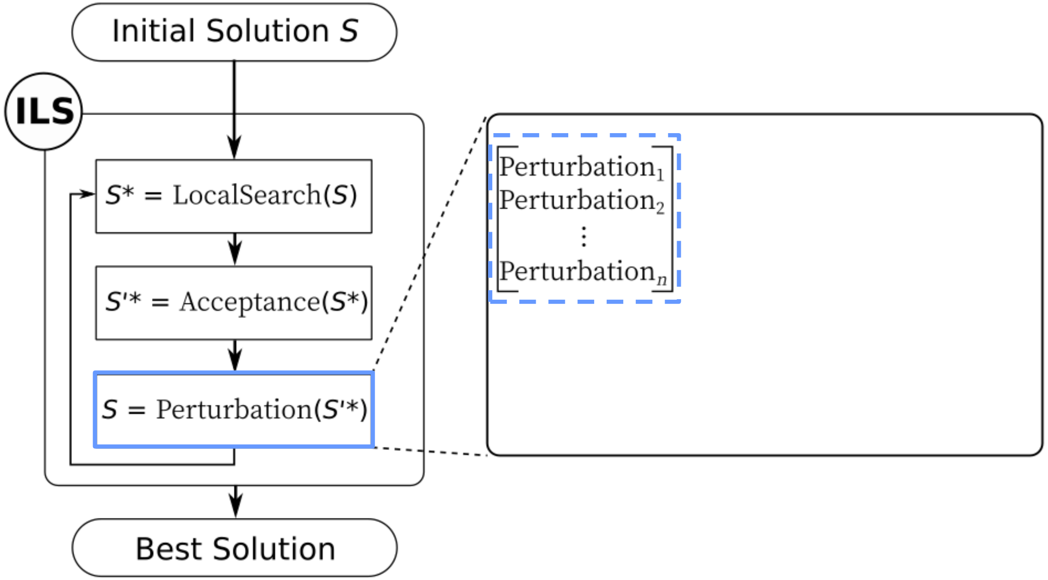


Proposed algorithm

Iterated local search with Multiple local search

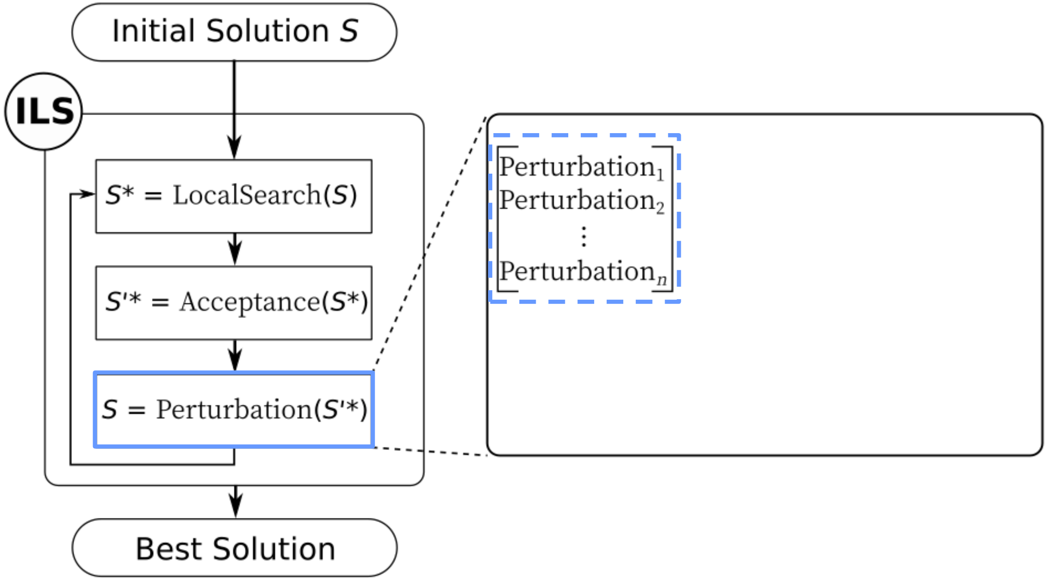


Iterated local search with Multiple perturbation



Proposed algorithm

Iterated local search with Multiple perturbation

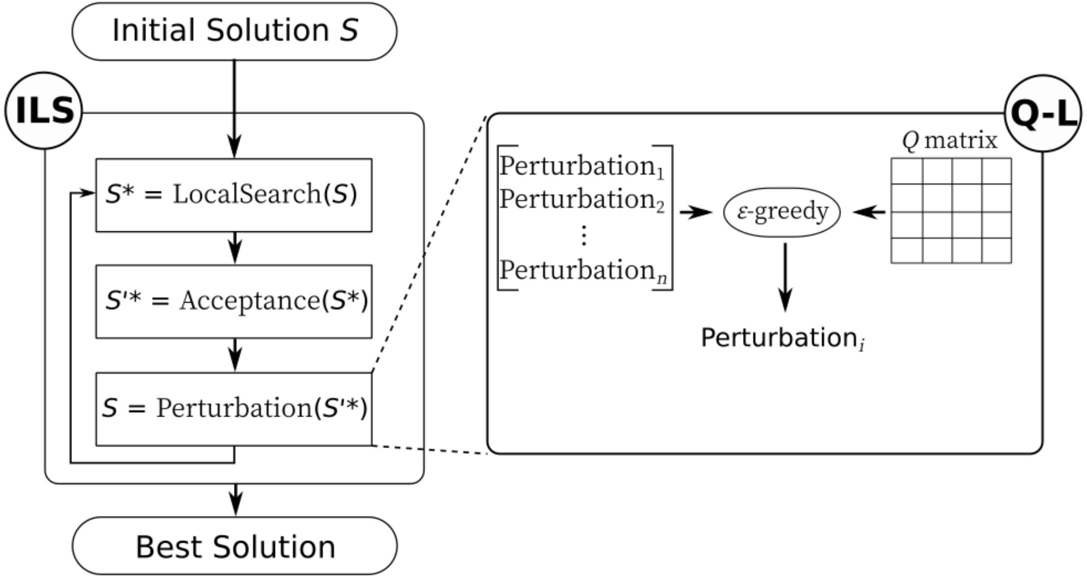


ILS

- **Perturbation operators**
  - Double-bridge
  - Shuffle-sequence
  - Reversion-sequence
- **Local search operator**
  - Developed 2opt

Proposed algorithm

Integration of Q-learning into Iterated local search



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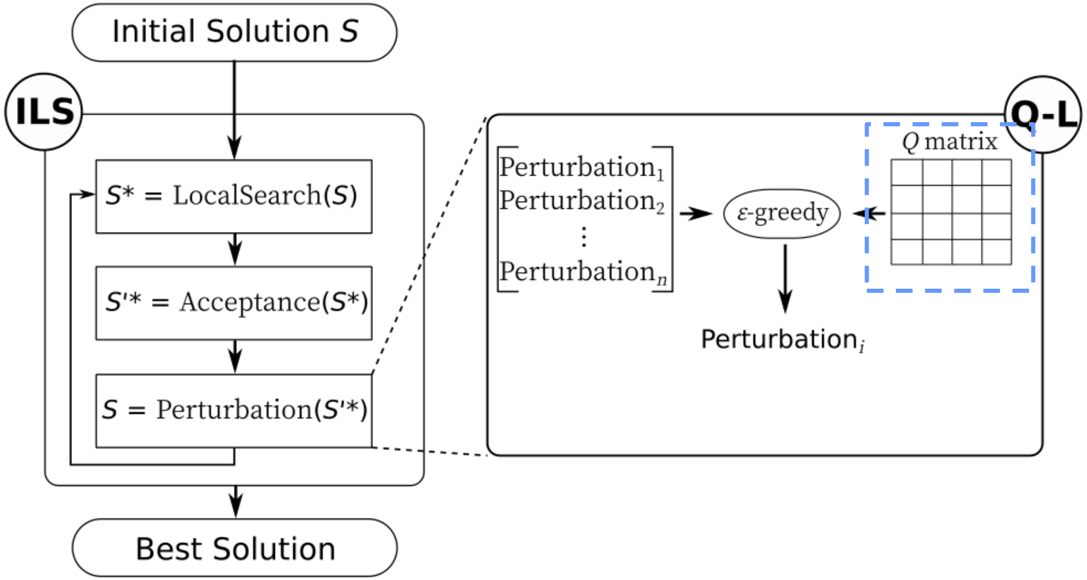

Q-L

- **State**

1: if an improvement occurs in the best found solution 0: otherwise

- **Action**

(perturbation type, repetition number)

- **Reward**

Improvement of objective function

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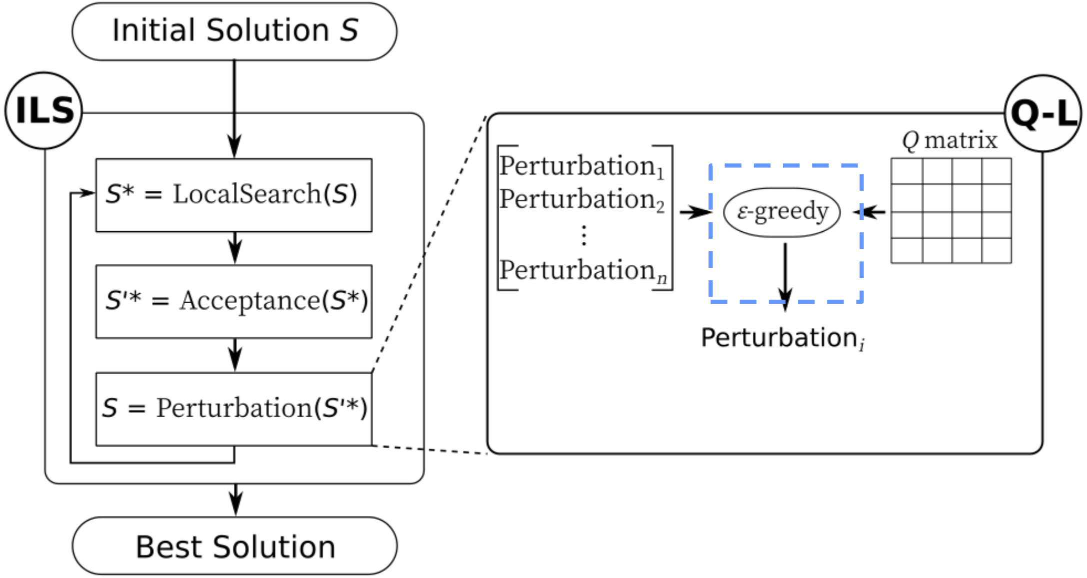
Proposed algorithm

Integration of Q-learning into Iterated local search



Q-L

- **State**

1: if an improvement occurs in the best found solution 0: otherwise

- **Action**

(perturbation type, repetition number)

- **Reward**

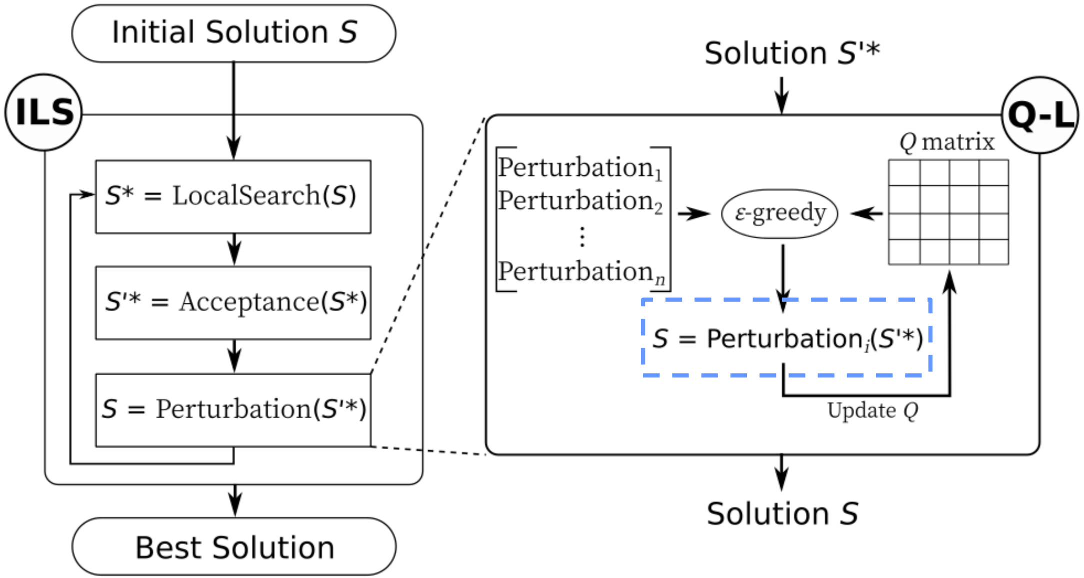
Improvement of objective function

- **Action selection**

ε-greedy strategy

Proposed algorithm

Integration of Q-learning into Iterated local search



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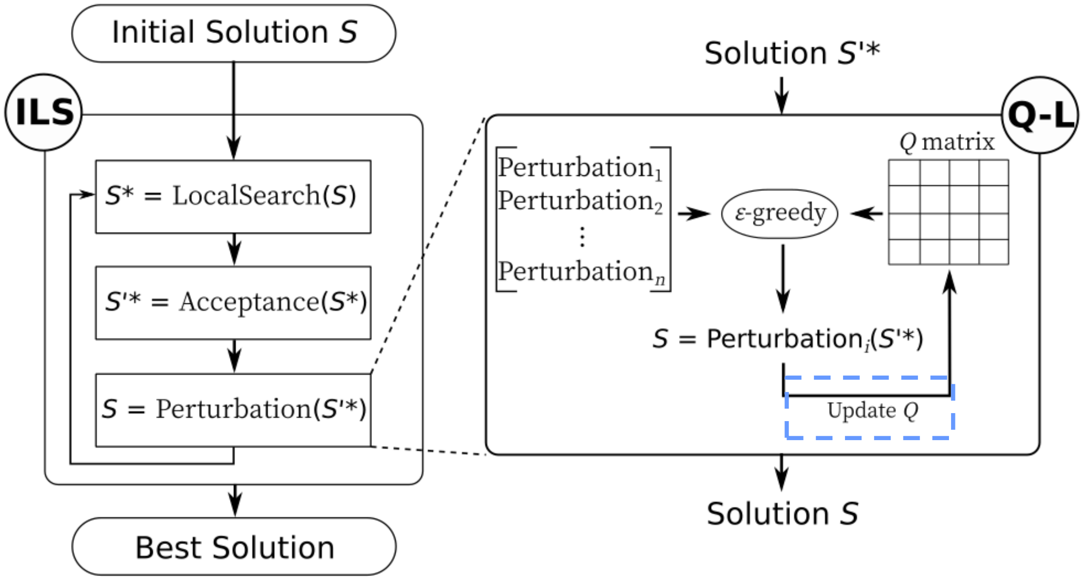
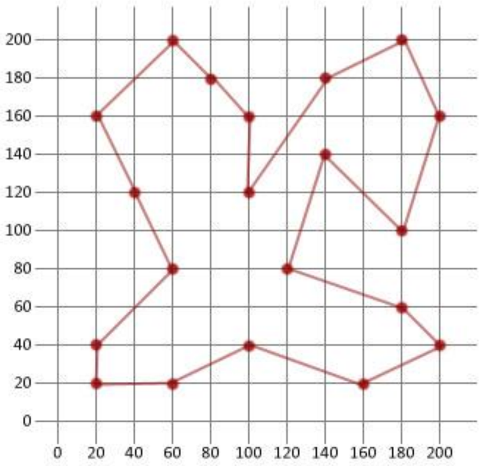


Experimental design

Traveling Salesman Problem (TSP)

- Data from TSPLIB
- Instances with 50-2000 cities



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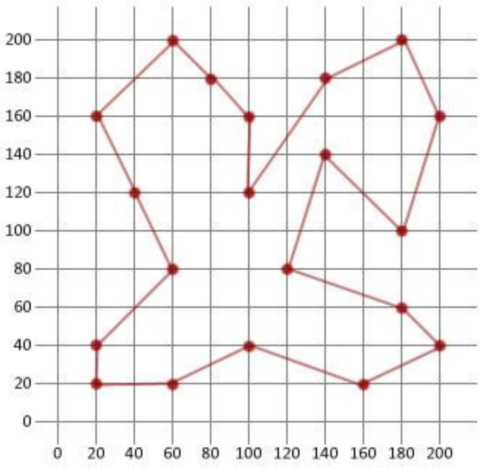

Parameter tuning (Design of Experiments)

- *Parameters:*
  - ϵ (*Epsilon)*
  - β (*Epsilon decay)*
  - α (*Learning rate)*
  - γ *(Discount factor)*

Experimental design

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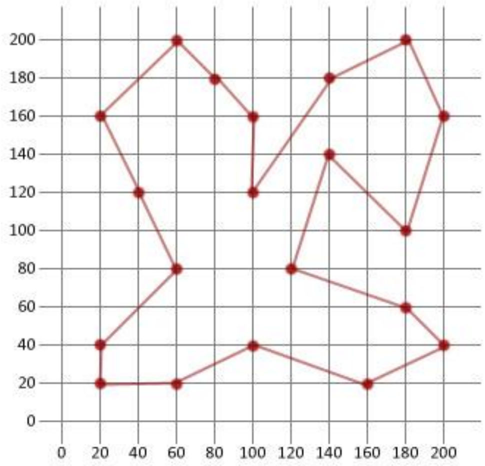

Parameter tuning (Design of Experiments)

- *Parameters:*
  - ϵ (*Epsilon*)
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Performance Criteria

- Quality of solutions (RPD)

$$RPD = \frac{OF - OF^*}{OF^*} \times 100$$

- Computational effort (time)
- Convergence behavior

Results

Performance criteria

- Quality of solutions (RPD%)
- Computational effort (time)

Instance	Optimal	Best RPD(%)	Best time (s)	Average RPD(%)	Average time (s)
berlin52	7542	0	0	0	0.2
st70	675	0	0.7	0.109	4.5
kroA100	21282	0	1.3	0	7.3
rd100	7910	0	3.6	0.201	14.2
lin105	14379	0	0.4	0	4.5
pr124	59030	0	1.8	0	8.7
ch130	6110	0	17.4	0.359	40.4
ch150	6528	0	19.8	0.374	33.2
u159	42080	0	9.6	0.112	42.9
d198	15780	0.057	149.4	0.180	147.5
kroA200	29368	0	117.3	0.150	194.8
ts225	126643	0	56.5	0.002	108.9
pr264	49135	0	61.4	0.172	135.8
a280	2579	0	246.4	0.498	276.5
pr299	48191	0	453.1	0.274	440.7
lin318	42029	0.302	502.4	0.795	479.6
fl417	11861	0.211	574.5	0.339	553.0
pr439	107217	0.438	573.9	1.392	528.2
pcb442	50778	0.640	518.4	1.294	521.6
d493	35002	0.923	571.5	1.499	587.8
vm1084	239297	3.061	2729.6	3.717	2608.3
d1291	50801	2.281	1985.9	3.418	2060.1
u1817	57201	3.449	412.1	4.366	1416
u2152	64253	3.947	1967.5	4.531	1022.3

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Results

Q1. effectiveness of multiple operators against single operator

- **Proposed algorithm** vs **Single ILS** (S-ILS)

Q2. effectiveness of using Q-learning for operator selection

- **Proposed algorithm** vs **Random ILS** (R-ILS)

- Performance criteria

$$RPD^{R(S)} = \frac{OF^{R(S)} - OF^{Q}}{OF^{Q}} \times 100$$

Instance	R-ILS	S-ILS		
		1	2	3
berlin52	3.34	3.48	3.17	2.10
st70	1.19	1.25	0.74	0.83
kroA100	0.97	1.23	0.58	0.64
rd100	1.43	1.79	1.20	1.62
lin105	0.83	1.84	1.24	0.85
pr124	1.06	1.21	0.65	0.70
ch130	1.68	2.18	1.12	1.23
ch150	1.31	2.36	1.01	1.17
u159	2.28	2.99	1.28	1.74
d198	0.73	1.92	0.71	0.96
kroA200	1.33	1.91	0.60	1.02
ts225	0.72	1.54	0.74	0.80
pr264	1.67	3.39	1.12	1.14
a280	2.23	4.17	1.53	2.00
pr299	2.22	4.23	1.65	2.18
lin318	1.81	3.34	1.44	1.79
fl417	1.14	3.88	2.19	1.99
pr439	2.78	3.65	1.75	1.45
pcb442	1.26	3.16	1.66	1.97
d493	1.26	3.58	2.18	2.06
vm1084	1.33	2.74	1.40	1.89
d1291	2.06	3.33	2.30	2.62
u1817	1.57	3.18	3.14	3.24
u2152	1.75	2.72	2.43	2.52

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$$RPD^{R(S)} = \frac{OF^{R(S)} - OF^{Q}}{OF^{Q}} \times 100$$

- Worst-case complexity
  - Proposed algorithm: $O(I(N^3 + A))$
  - R-ILS: $O(I(N^3))$

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$$RPD^{R(S)} = \frac{OF^{R(S)} - OF^{Q}}{OF^{Q}} \times 100$$

- Worst-case complexity
  - Proposed algorithm: $O(I(N^3 + A))$
  - R-ILS: $O(I(N^3))$
  - Overhead of Q-learning: $O(I \times A)$

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Conclusion

- Proposed algorithms vs S-ILS and R-ILS
  - ILS with **Multiple perturbation operators** outperforms single operator
  - Integration of **Q-learning** for operator selection improves the performance
  - No significant extra complexity overhead
  - Q-learning is very useful when the **operators are competitive**

Perspective

- Solving COPs with several competitive operators

Thanks for your attention

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